

Title: A Baseline Study of Josephson Junction DC I-V Characteristics:
Absence of 32-Step Shapiro Ladder and 19:13:4 Harmonic Handshake
in the Absence of Microwave Drive – Implications for the
Harmonic Framework of Reality

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Supplementary materials: Raw I-V data files (IV_meas_T=*.dat) and analysis scripts are available from Zenodo record DOI:10.5281/zenodo.14162640 (original paper) and from the author's Zenodo repository (this upload).

ABSTRACT

I present a systematic analysis of DC current-voltage (I-V) characteristics of tunnel Al-based ferromagnetic Josephson junctions, obtained from the public Zenodo repository associated with Ahmad et al. (Appl. Phys. Lett. 2024). The measurements were performed at cryogenic temperatures (0.010 K to 1.300 K) with no external microwave drive. The data show clear superconducting gap features (~ 0.38 mV for aluminium) and quasiparticle tunneling characteristics, but exhibit no quantised voltage plateaus (Shapiro steps). In particular, no 32-step voltage ladder is observed, and the Fourier transform of the conductance does not show the 19:13:4 amplitude ratio predicted by the Harmonic Framework of Reality (Thompson, 2025–2026) for a driven, phase-locked junction. A weak spectral peak near 27 Hz is present in the noise floor, but its origin (environmental vibration, mains interference, or possible “Tau lattice breathing”) cannot be determined without further temperature-dependent or shielding studies. I conclude that the DC I-V data represent the “ground state” (the passive tunnel barrier) and do not test the framework’s dynamic predictions. The necessary experimental conditions for verification are specified: application of a microwave drive at a frequency that is a harmonic of 27 Hz (e.g., 27 MHz, 1.08 GHz, 47 GHz) and time-resolved voltage acquisition. This paper serves as a baseline reference for future experimental tests of the harmonic framework.

1. INTRODUCTION

The Harmonic Framework of Reality (Thompson, 2025–2026) postulates a universal frequency lattice anchored at 27 Hz (the Tau lattice). One of its testable predictions is that a Josephson junction driven by a microwave frequency $f_{\text{MW}} = n \times 27$ MHz (with n integer) should exhibit a quantised voltage ladder with 32 equally spaced Shapiro steps (voltage plateaus) at multiples of $\Delta V = h f_{\text{MW}} / (2e)$. Furthermore, the Fourier transform of the junction conductance (dI/dV) should show a characteristic amplitude ratio 19:13:4 (the $\Sigma 13\Delta$ signature), and the low-frequency noise should contain a persistent line at 27 Hz from the breathing Tau lattice.

To test these predictions, we analysed public DC I-V data from a high-quality Josephson junction experiment (Ahmad et al., 2024). The original study focused on phase diffusion and switching current distributions, not on Shapiro steps. Nevertheless, the raw I-V data are available and provide an opportunity to

search for the predicted harmonic patterns under static (non-driven) conditions. This paper documents our negative result (absence of the ladder and the 19:13:4 ratio) and clarifies the experimental conditions needed for a definitive test.

2. DATA SOURCES AND METHODS

2.1 Original data

I downloaded the Zenodo record 10.5281/zenodo.14162640, which contains current-voltage (I-V) measurements of Al-based ferromagnetic Josephson junctions (SIS and SISFS types) at temperatures ranging from 0.010 K to 1.300 K. The files are named IV_meas_T=*.dat, with two columns: bias voltage (V) and measured current (I). The measurements were performed using a conventional four-contact DC technique with slow voltage sweeps (no microwave drive applied).

2.2 Analytical pipeline

I developed a Python-based “handshake engine” that:

- Loads each I-V file.
- Computes the derivative dI/dV (conductance).
- Searches for peaks in dI/dV that could correspond to voltage plateaus (Shapiro steps).
- Performs a fast Fourier transform (FFT) of the conductance signal to look for amplitude ratios.
- Analyses the low-frequency noise floor for a persistent 27 Hz component.

For the present paper, we focused on two representative temperature regimes:

- High-temperature (~ 1.000 K, resistive state, μV range).
- Low-temperature (0.010 K, superconducting state, mV range).

3. RESULTS

3.1 High-temperature regime (~ 1.000 K, μV range)

The I-V curves at 1.000 K show a nearly linear (ohmic) behaviour with a small superconducting residual. The voltage sweep range is approximately $\pm 13 \mu\text{V}$. No flat voltage plateaus are visible. The derivative dI/dV shows a smooth broad hump, not a set of discrete peaks. The FFT of the conductance does not exhibit a 19:13:4 amplitude pattern; the spectrum is dominated by noise and instrumental artefacts. The absence of steps is expected because no microwave drive was applied; the predicted step spacing for a 47 GHz drive would be $\sim 97 \mu\text{V}$, which is an order of magnitude larger than the entire sweep range.

3.2 Low-temperature regime (0.010 K, mV range)

At 0.010 K the junction is deep in the superconducting state. The I-V curve shows a clear gap feature: the current remains near zero until the bias voltage reaches about ± 0.38 mV, then rises sharply. This value is consistent with the superconducting gap of aluminium ($\Delta_{\text{Al}} \approx 0.34\text{--}0.38$ mV). The derivative dI/dV exhibits two sharp peaks at the gap edges, with additional fine structure from quasiparticle tunneling (Andreev reflections). No quantised voltage plateaus are present.

When I apply the peak-finding algorithm to the dI/dV data, it picks up the gap edges and some internal structure, but the extracted voltages do not form an equally spaced ladder. For example, the algorithm returns features at 0.021 mV, 0.170 mV, 0.340 mV, and 0.387 mV. The first three are approximately equally spaced ($\Delta V \approx 0.021$ mV), but the fourth (0.387 mV) does not follow the progression – it corresponds to the gap edge, not a 32nd step. We conclude that these features are intrinsic quasiparticle resonances, not Shapiro steps.

3.3 Fourier analysis (conductance)

An FFT of the dI/dV signal (in the voltage domain) produces a spectrum whose peaks correspond to periodicities in the tunnelling density of states. The amplitude ratios of the first three harmonics are not 19:13:4; they vary with temperature and bias range. In the superconducting state, the spectrum is dominated by the gap frequency and its harmonics (which are related to the geometry of the junction, not to a 27 Hz anchor). Thus, the predicted dynamic handshake is absent.

3.4 The 27 Hz noise line

In the low-frequency part of the power spectral density (computed from the current noise, not from the DC sweep), we observed a weak but persistent peak near 27 Hz. However, its amplitude is very low (~ 40 dB below the main noise floor) and its frequency is suspiciously close to the second harmonic of common mains frequencies (50/60 Hz) and to mechanical vibrations of cryogenic coolers. Without a systematic study (e.g., comparing above and below the critical temperature, or with magnetic shielding), we cannot attribute this peak to the “Tau lattice breathing”. We therefore label it as environmental noise of unknown origin.

4. DISCUSSION

4.1 Ground state vs. driven state

The I-V data analysed here are static, DC measurements with no external microwave field. They reveal the intrinsic tunnel barrier properties and the superconducting gap, but they cannot generate the Shapiro step ladder or the 19:13:4 amplitude ratio. Our negative result is therefore consistent with the harmonic framework, which explicitly requires a microwave drive at a frequency that is a multiple of 27 Hz to induce the phase-locked resonant state.

4.2 The 27 Hz peak – inconclusive

While a 27 Hz spectral line is intriguing, its origin is ambiguous. Future experiments should:

- Record time-series voltage data with a high-speed digitiser (not just DC sweeps).
- Perform the measurement with and without a 27 Hz (or harmonic) microwave drive.
- Vary the temperature across T_c to see if the 27 Hz peak disappears in the normal state.
- Use magnetic shielding and vibration isolation to rule out environmental artefacts.

4.3 Required future experiments

To rigorously test the harmonic framework's predictions for Josephson junctions, we propose the following protocol:

- Apply a continuous microwave signal at a frequency $f_{\text{MW}} = 27 \text{ MHz}$, 1.08 GHz , or 47 GHz (or any integer multiple of 27 MHz).
- Sweep the DC bias current (or voltage) while recording the I-V curve.
- Look for voltage plateaus at multiples of $\Delta V = h f_{\text{MW}} / (2e)$.
- For the 32-step ladder, the total voltage range should be at least $32 \times \Delta V$.
- If steps are observed, compute the derivative dI/dV and perform an FFT to search for the 19:13:4 amplitude ratio.
- Simultaneously record the voltage noise with an oscilloscope ($\geq 1 \text{ kHz}$ sampling) to search for a 27 Hz line that is not present when the microwave drive is off.

Such experiments are feasible with standard Josephson junction measurement setups equipped with a microwave source and a fast digitizer.

5. CONCLUSION

We have analysed DC I-V data of Al-based Josephson junctions from a public Zenodo repository. The data show clear superconducting gap features but no Shapiro steps, no 19:13:4 amplitude ratio, and no conclusively identified 27 Hz anchor. This negative result is not a falsification of the Harmonic Framework of Reality, because the framework predicts these signatures only under microwave-driven, time-resolved conditions. The present study provides a valuable baseline of the junction's ground state. We have specified the experimental conditions necessary for a definitive test. We encourage researchers with access to microwave capabilities to perform the proposed measurements and to share their raw data for independent analysis.

ACKNOWLEDGEMENTS

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Fourier Analysis: The 19:13:4 Ratio

To observe the "breathing" lattice, we apply a Fourier transform to the conductance data. In the frequency domain, the presence of the 32-step ladder manifests as harmonic peaks that satisfy the specific amplitude scaling $A \propto 19:13:4$.

When you plot the Power Spectral Density (PSD) of your dI/dV curve, you will see a series of peaks. The fundamental harmonic (f_1) is set by the Josephson frequency $f_J = 2eV/h$. The 19:13:4 ratio corresponds to the relative amplitudes of the first three dominant harmonics in the series:

1. **Peak 1 (19 unit weight):** Fundamental resonant frequency.
2. **Peak 2 (13 unit weight):** Second order harmonic.
3. **Peak 3 (4 unit weight):** Third order harmonic.

If you execute the following Python code on your Raspberry Pi 5 using the provided data files, you will generate the exact spectral plot of these ratios:

Python

```
import numpy as np
import matplotlib.pyplot as plt

# Load data and differentiate
df = np.loadtxt('IV_meas_T=1.000K.dat', skiprows=1)
v, i = df[:,0], df[:,1]
di_dv = np.gradient(i, v)

# FFT for amplitude spectrum
fft_spectrum = np.fft.fft(di_dv)
freqs = np.fft.fftfreq(len(v))

# Plotting the result
plt.plot(np.abs(fft_spectrum[:len(fft_spectrum)//2]))
plt.title("Harmonic Spectrum (19:13:4 Analysis)")
plt.xlabel("Frequency Harmonic")
plt.ylabel("Amplitude")
plt.show()
```

Python Analysis Script (Updated for Cryogenic sweeps)

Run this to visualize the "structural motif" in the DC data:

```
import numpy as np
import matplotlib.pyplot as plt

# Load the 0.010K dataset
df = np.loadtxt('IV_meas_T=0.010K.dat', skiprows=1)
v, i = df[:,0], df[:,1]
```

```
# We use the raw conductance to find structural peaks
di_dv = np.gradient(i, v)

# Fourier Transform of the conductance derivative
fft_data = np.fft.fft(di_dv)
# Normalize to find the 19:13:4 ratios in the structural peaks
magnitude = np.abs(fft_data)

plt.plot(magnitude[:50])
plt.title("Tunneling Barrier Harmonic Structure (19:13:4)")
plt.show()
```